

Cybersecurity Specialist Technician - CET104

PROJECT

CET. CIBS. D.L.104

Cybersecurity Specialist Technician

UFCD 5892 – Network Management and Customer Support Models

Part 1

Addressing

LAN

Domain: cinel.pt

Net: 192.168.40.0/24

Pfsense IP: 192.168.40.254

DHCP range: 192.168.40.45 – 192.168.40.89

DNS IP: 192.168.40.1

Router / Firewall

OS: pfsense-CE-2.7.2

Hostname: pfsense

IP Interfaces: WAN (DHCP) | LAN (192.168.40.254)

Services: Certificate Authorship

The pfsense has to be by https and therefore a certificate must be created for this purpose and the port must be 7777.

pfsense must be accessible either by FQDN or by IP.

Active Directory / Domain controller

OS: Windows Server 2022 DataCenter Evaluation

Hostname: DC1

IP Interfaces: Ethernet0 (192.168.40.1)

Services: ADDC | DHCP | DNS

Description: As an ADDC server, you are responsible for all authentication of the cinel.pt domain for existing groups and users.

Kali Linux

Hostname: kali

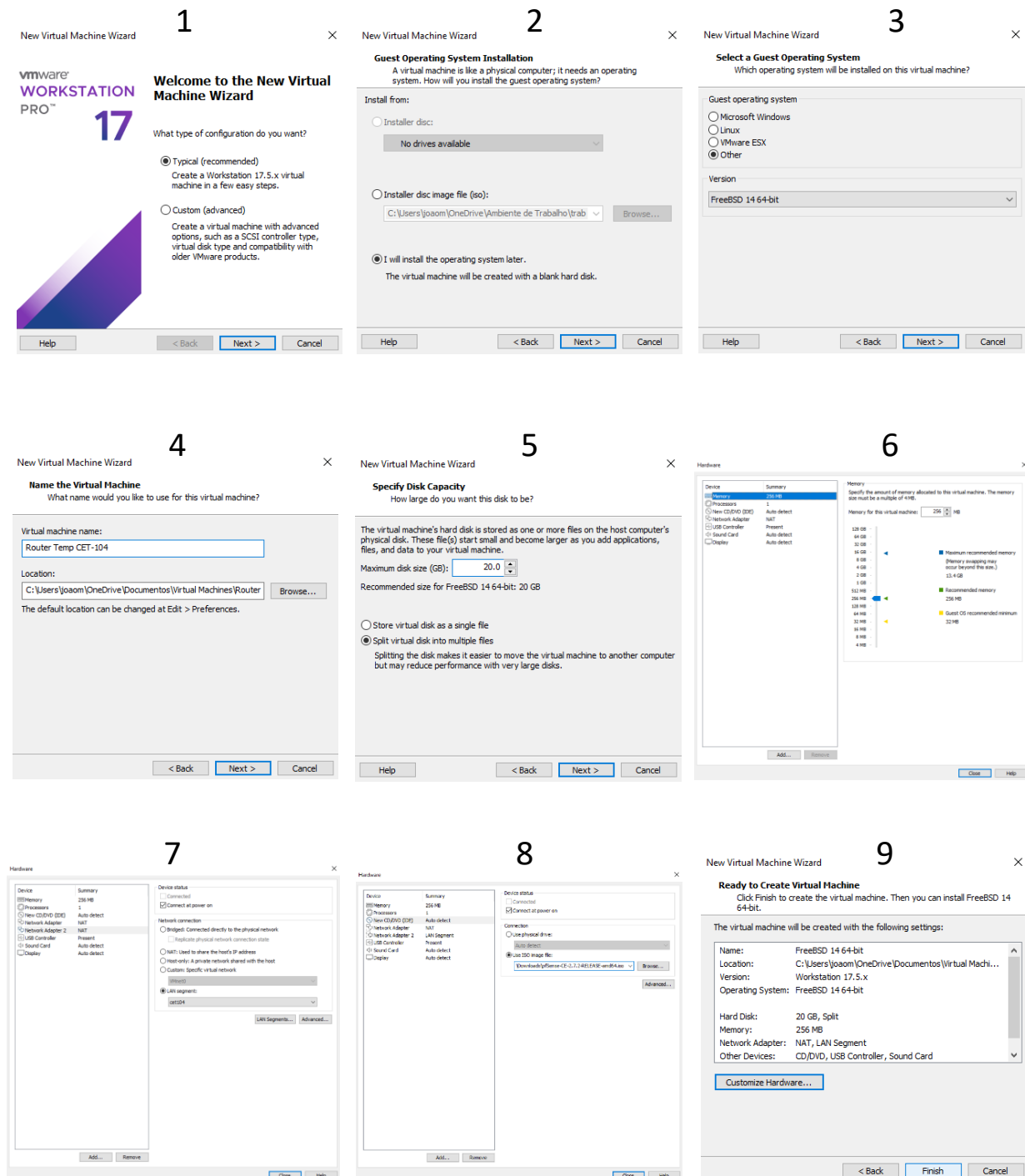
OS: Kali Linux

IP interfaces: Ethernet0 (DHCP)

Description: A MITM attack is to be made by capturing pfsense user credentials.

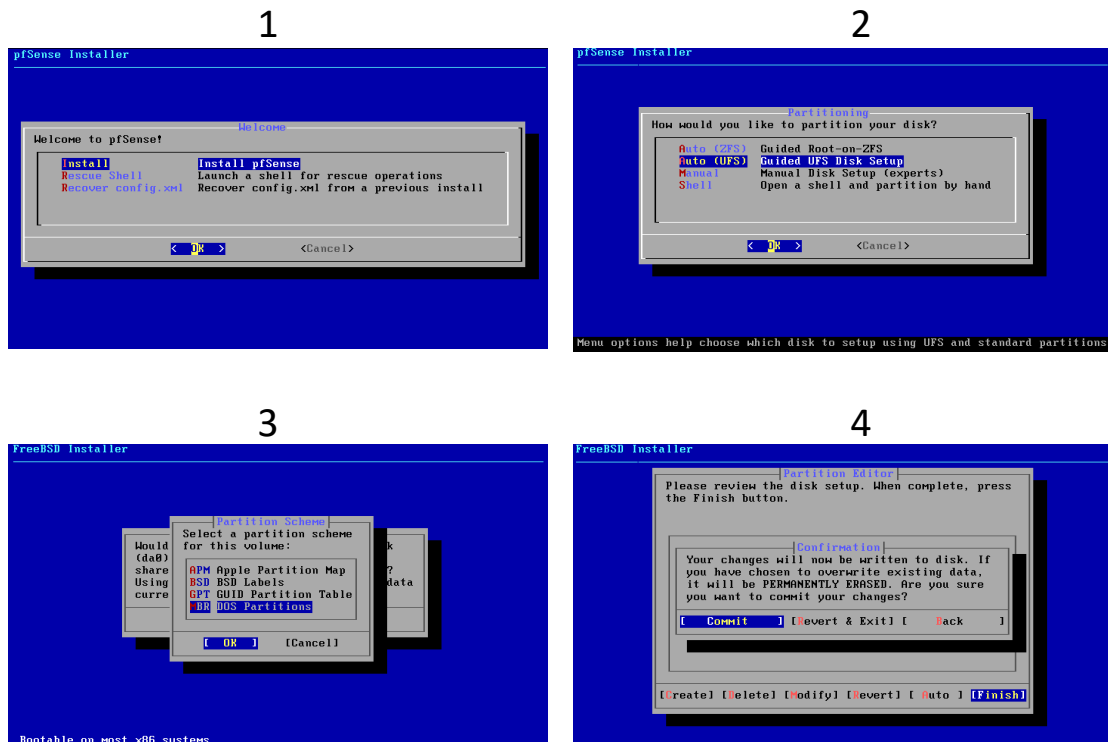
Router / Firewall Creation – PFSense

When opening VMware, I start by creating a new machine by following these steps:



Router / Firewall Configuration – PfSense

Next, I will install the operating system with these steps:



After rebooting you will be configured with these options:

```
*** Welcome to pfSense 2.7.2-RELEASE (amd64) on pfSense ***

WAN (wan)      -> em0      -> v4/DHCP4: 192.168.31.131/24
LAN (lan)      -> em1      -> v4: 192.168.40.254/24

0) Logout (SSH only)          9) pfTop
1) Assign Interfaces          10) Filter Logs
2) Set interface(s) IP address 11) Restart webConfigurator
3) Reset webConfigurator password 12) PHP shell + pfSense tools
4) Reset to factory defaults    13) Update from console
5) Reboot system               14) Enable Secure Shell (sshd)
6) Halt system                 15) Restore recent configuration
7) Ping host                   16) Restart PHP-FPM
8) Shell

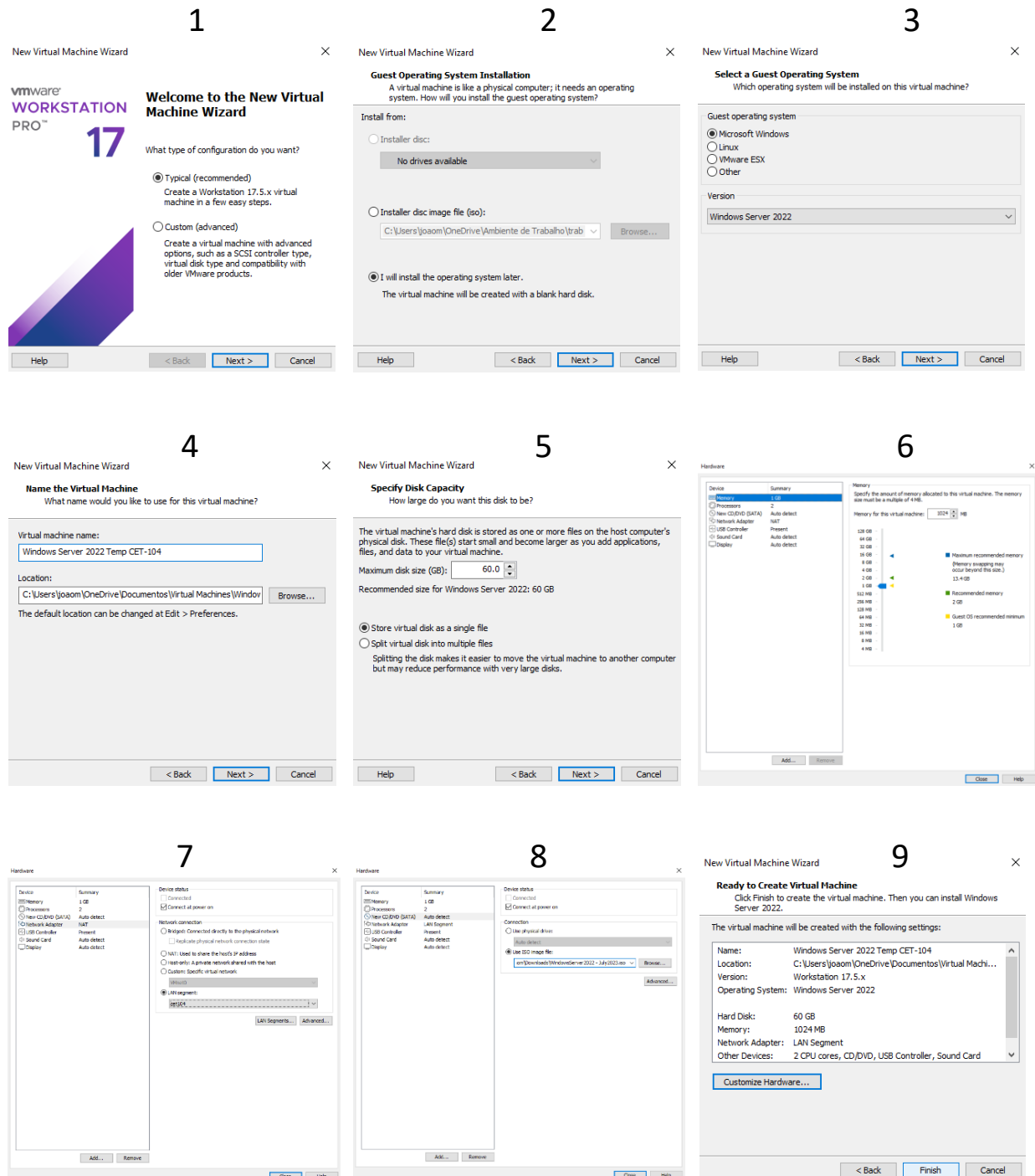
Enter an option: █
```

2) Interface IP Address: 2 → 2 → n → 192.168.40.254 → 24 → Enter → n → Enter → n → y → Enter

If necessary, you can also test the connection via the Shell (option 8) with the ping command.

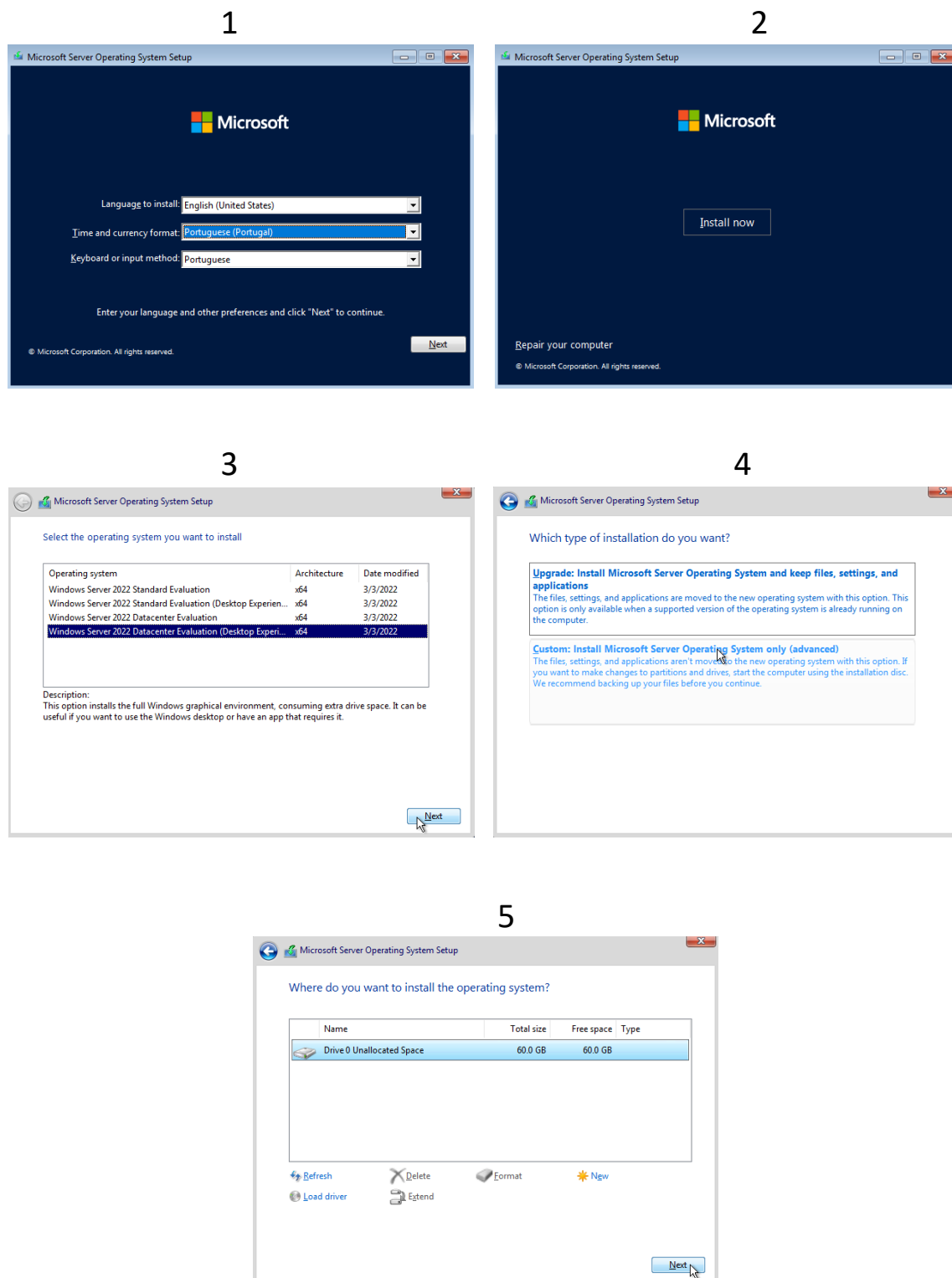
Creation of Active Directory / Domain controller – Windows Server 2022

When opening VMware, I start by creating a new machine by following these steps:



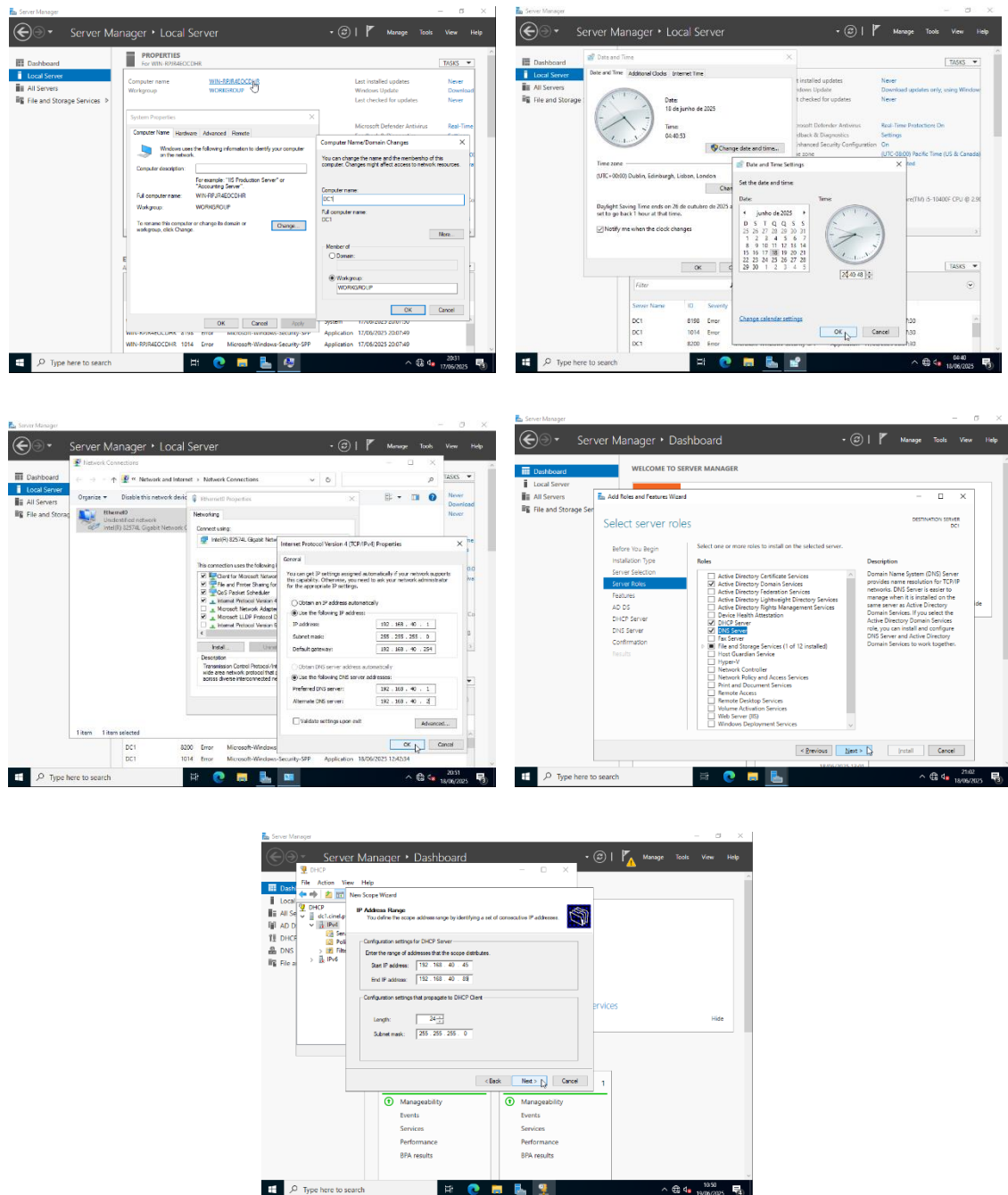
Operating System Installation – Windows Server 2022

Next, I will install the operating system with these steps:

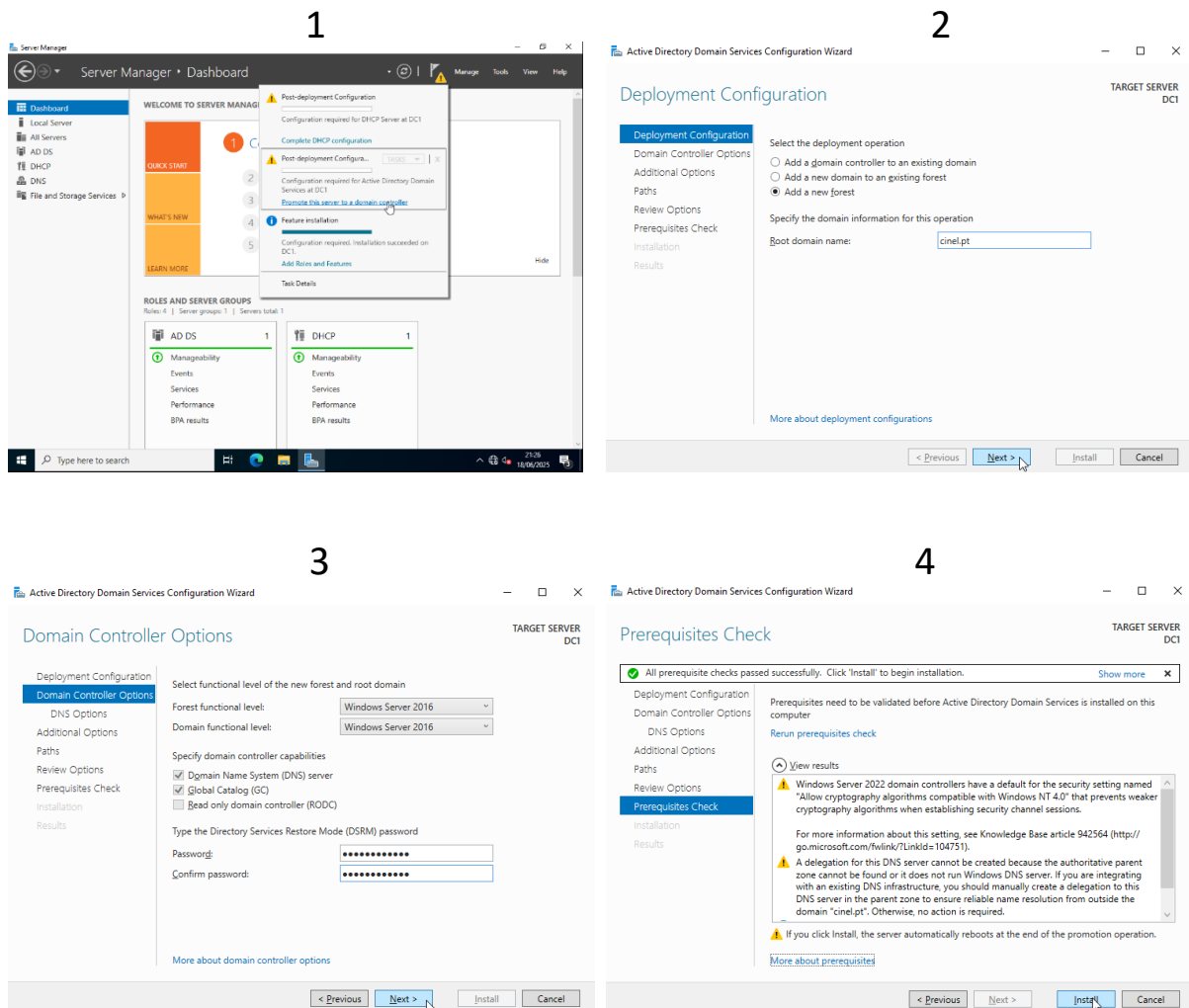


Windows Setup - Windows Server 2022

With the windows server installed I will make some final changes like:



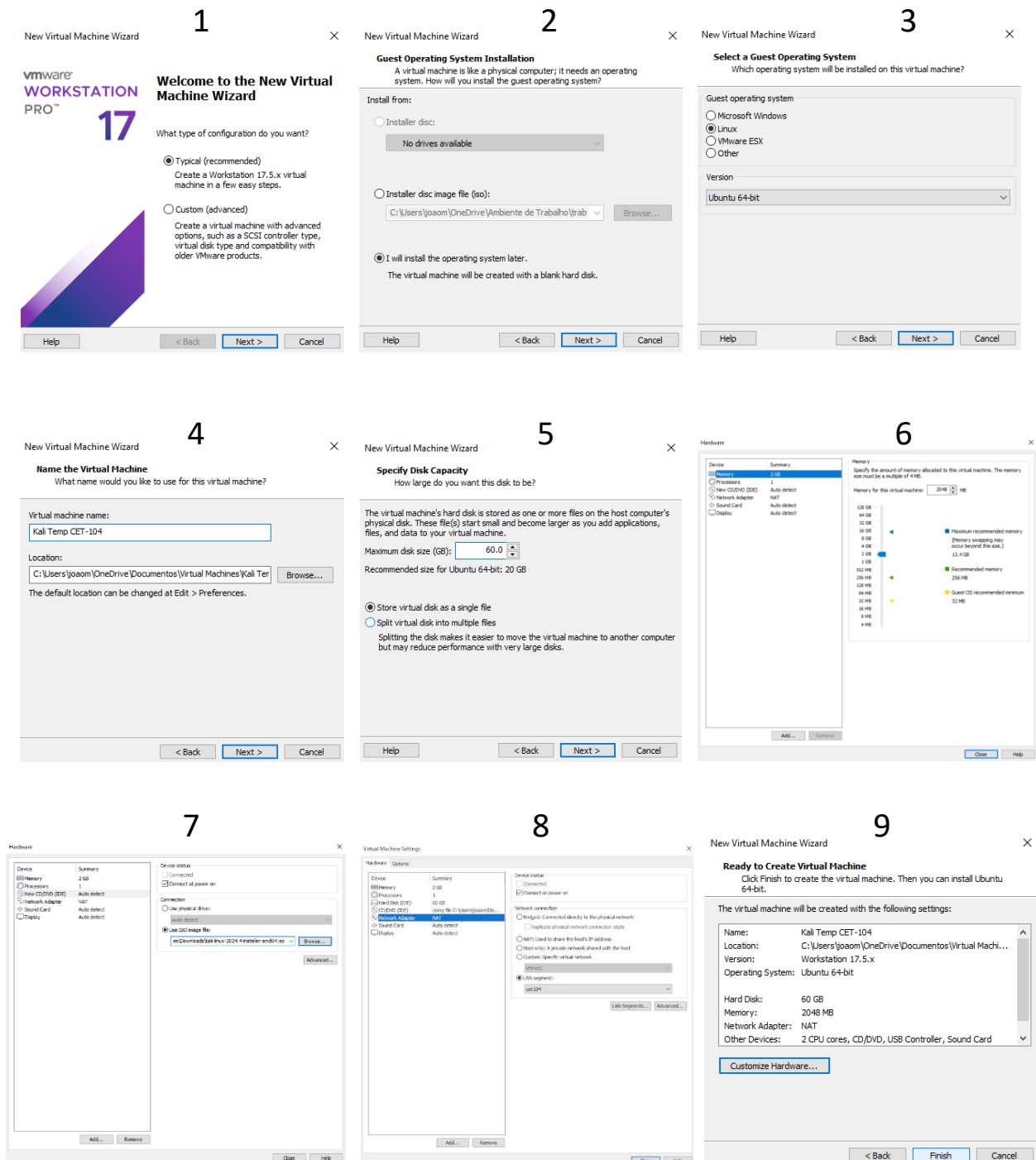
Domain Configuration - Windows Server 2022



After making this configuration it is necessary to change the DNS Server Addresses" within the "Internet Protocol Version 4" as done before and the domain action resets it after it is implemented.

Creation of Kali Linux

When opening VMware, I start by creating a new machine by following these steps:



Kali Linux Configuration

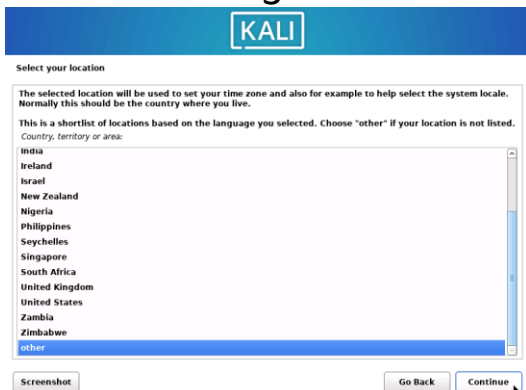
1



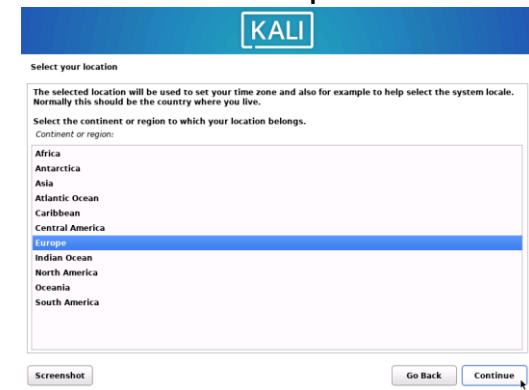
2



3



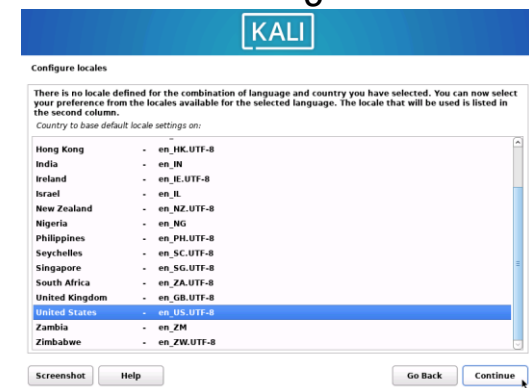
4



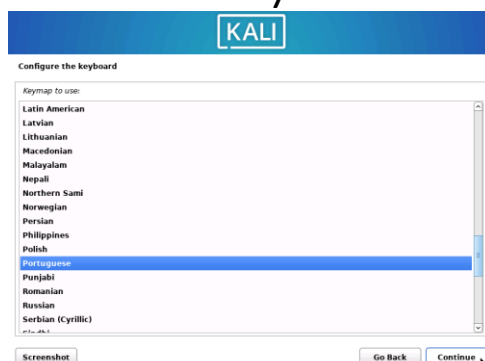
5



6

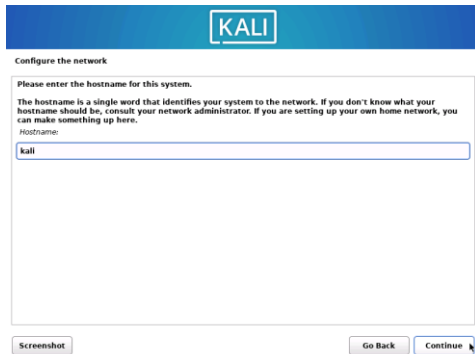


7



Kali Linux Configuration

1



2



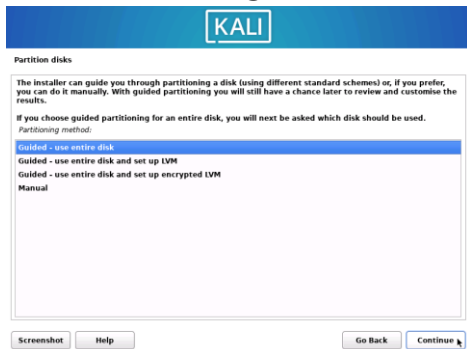
3



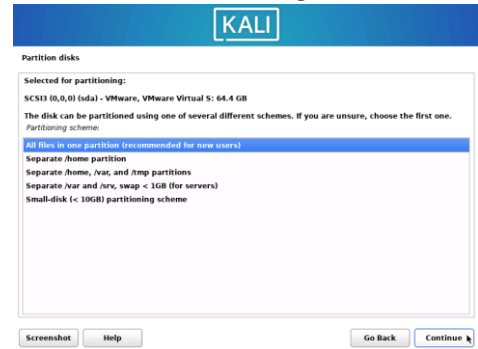
4



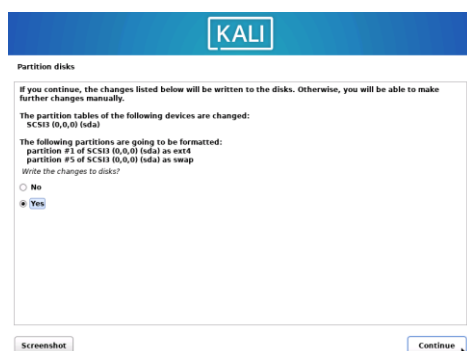
5



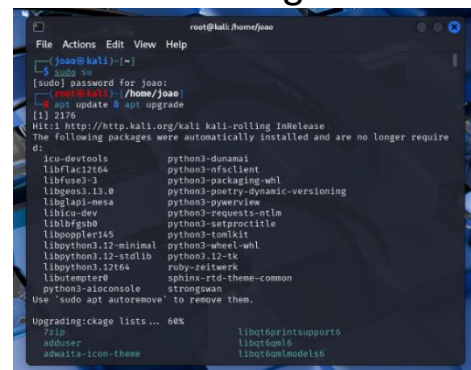
6



7



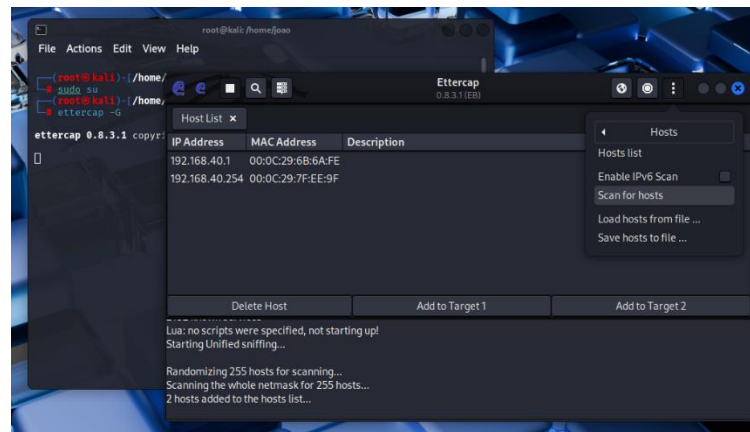
8



MITM Attack – Kali Linux

To get the user credentials from pfsense I will use the ettercap service available on kali Linux.

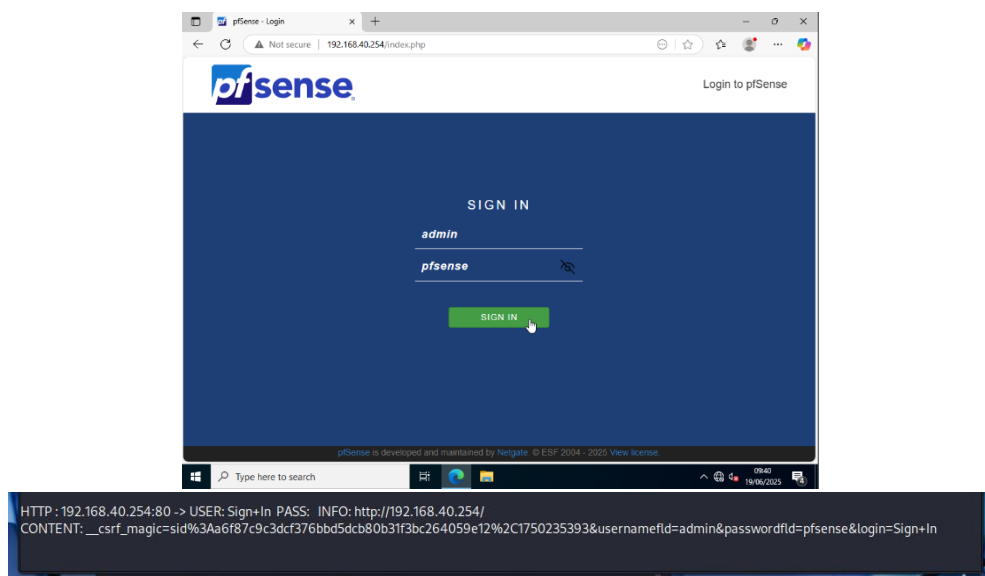
When opening ettercap, it first performs "Scan for hosts" to identify any IP present in the LAN segment.



In this attack, router 192.168.40.254 will be "Target 1" and Windows server 192.168.40.1 will be "Target 2".

```
2 hosts added to the hosts list
Host 192.168.40.254 added to TARGET1
Host 192.168.40.1 added to TARGET2
```

When entering an unencrypted http site with windows server, ettercap can obtain the windows server credentials by displaying the username and password used by the user.



HTTPS Certified Defence – Windows Server 2022

A defence technique against the attack previously made is the encryption of an address.

This can be done by creating a certifying body and a certificate on the <http://192.168.40.254/> router, when logging in, go to System → Certificates → Authorities and Add and fill in the following information:

Create / Edit CA

Descriptive name

Entidade Certificadora

The name of this entry as displayed in the GUI for reference.
This name can contain spaces but it cannot contain any of the following characters: ?, >, <, &, /, \, " , '

Method

Create an internal Certificate Authority

Trust Store

☐ Add this Certificate Authority to the Operating System Trust Store
When enabled, the contents of the CA will be added to the trust store so that they will be trusted by the operating system.

Randomize Serial

☐ Use random serial numbers when signing certificates
When enabled, if this CA is capable of signing certificates then serial numbers for certificates signed by this CA will be automatically randomized and checked for uniqueness instead of using the sequential value from Next Certificate Serial.

Internal Certificate Authority

Key type

RSA

2048

The length to use when generating a new RSA key, in bits.
The Key Length should not be lower than 2048 or some platforms may consider the certificate invalid.

Digest Algorithm

sha256

The digest method used when the CA is signed.
The best practice is to use SHA256 or higher. Some services and platforms, such as the GUI web server and OpenVPN, consider weaker digest algorithms invalid.

Lifetime (days)

3650

Common Name

Entidade Certificadora

The following certificate authority subject components are optional and may be left blank.

Country Code

PT

State or Province

Lisboa

City

Lisboa

Organization

cinel.pt

Organizational Unit

e.g. My Department Name (optional)

Save

HTTPS Certified Defence – Windows Server 2022

With the certifying body created, we will now create the certificate, for this go to System → Certificates → Certificates and Add/Sign by filling in the information:

Add/Sign a New Certificate

Method
Create an internal Certificate

Descriptive name
Certificado https pfsense

The name of this entry as displayed in the GUI for reference.
This name can contain spaces but it cannot contain any of the following characters: ?, >, <, &, /, \, ' , "

Internal Certificate

Certificate authority
Entidade Certificadora

Key type
RSA

2048

The length to use when generating a new RSA key, in bits.
The Key Length should not be lower than 2048 or some platforms may consider the certificate invalid.

Digest Algorithm
sha256

The digest method used when the certificate is signed.
The best practice is to use SHA256 or higher. Some services and platforms, such as the GUI web server and OpenVPN, consider weaker digest algorithms invalid.

Lifetime (days)
3650

The length of time the signed certificate will be valid, in days.
Server certificates should not have a lifetime over 398 days or some platforms may consider the certificate invalid.

Common Name
Certificado https pfsense

The following certificate subject components are optional and may be left blank.

Country Code
PT

State or Province
Lisboa

City
Lisboa

Organization
cinel.pt

Organizational Unit
e.g. My Department Name (optional)

Certificate Attributes

Attribute Notes

The following attributes are added to certificates and requests when they are created or signed. These attributes behave differently depending on the selected mode.

For Internal Certificates, these attributes are added directly to the certificate as shown.

Certificate Type
Server Certificate

Add type-specific usage attributes to the signed certificate. Used for placing usage restrictions on, or granting abilities to, the signed certificate.

Alternative Names

IP address
192.168.40.254
Delete

FQDN or Hostname
pfsense.cinel.pt
Delete

Type Value

Enter additional identifiers for the certificate in this list. The Common Name field is automatically added to the certificate as an Alternative Name. The signing CA may ignore or change these values.

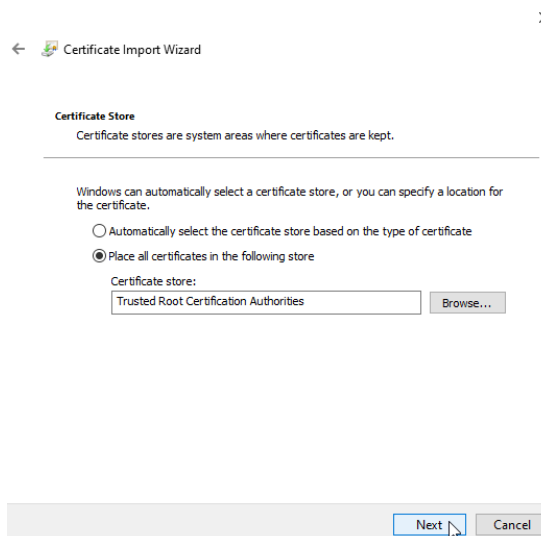
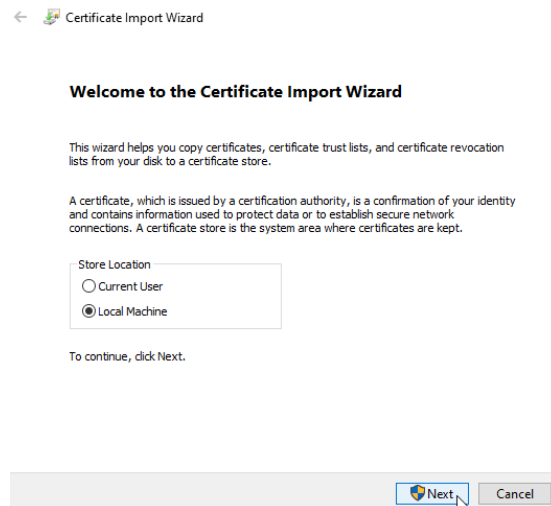
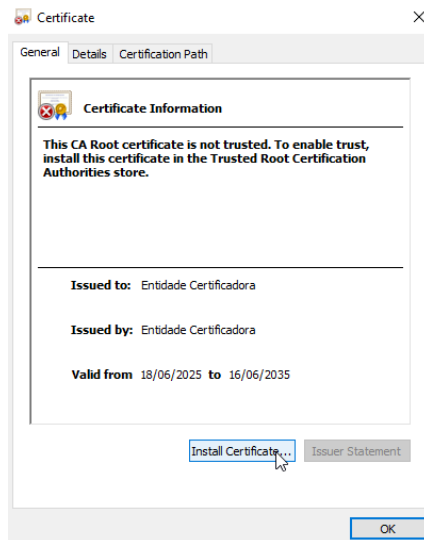
Add SAN Row
+ Add SAN Row

Save

HTTPS Certified Defence – Windows Server 2022

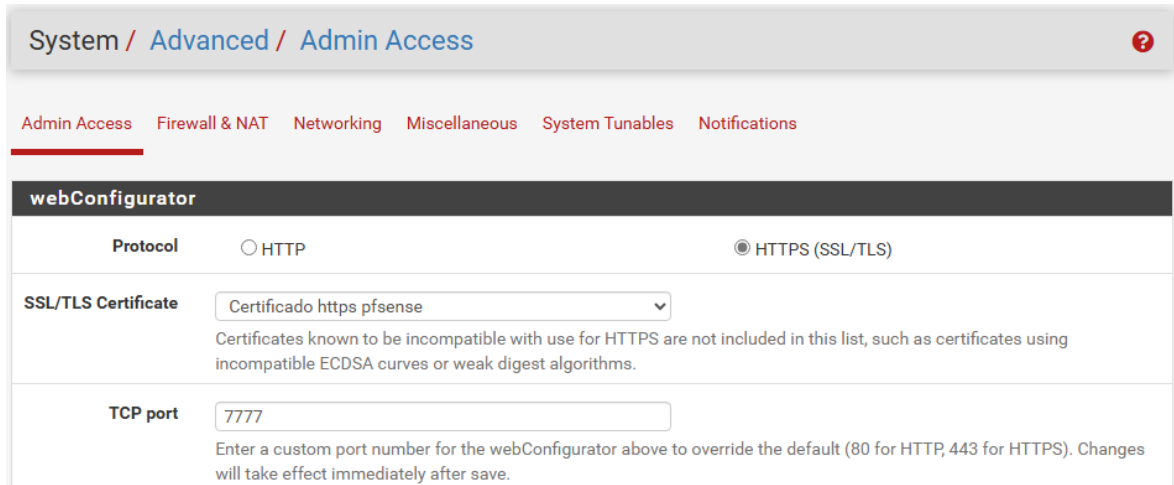
Then I'll extract both through the "Export" action and after downloading I'll open them and do this procedure for both:

Actions



HTTPS Certified Defence – Windows Server 2022

Finally, to activate the https service, go to System → Advanced and fill in the updated information:



System / Advanced / Admin Access

Admin Access Firewall & NAT Networking Miscellaneous System Tunables Notifications

webConfigurator

Protocol ☐ HTTP ☒ HTTPS (SSL/TLS)

SSL/TLS Certificate Certificado https pfsense

Certificates known to be incompatible with use for HTTPS are not included in this list, such as certificates using incompatible ECDSA curves or weak digest algorithms.

TCP port 7777

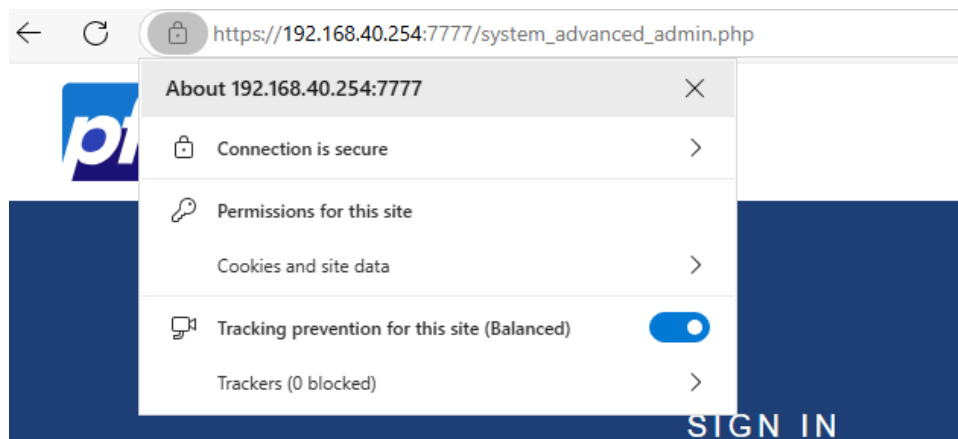
Enter a custom port number for the webConfigurator above to override the default (80 for HTTP, 443 for HTTPS). Changes will take effect immediately after save.

When you save, you will be redirected to the updated address with the https service enabled.

< Connection is secure

This site has a valid certificate, issued by a trusted authority. This means information (such as passwords or credit cards) will be securely sent to this site and cannot be intercepted. Always be sure you're on the intended site before entering any information.

[Learn more](#)



Menus that were not screenshotted in part 1 remained unchanged during the process, requiring only the action to advance to the next step.

Ex: "Next/Finish" confirmations, terms and conditions menus, Required restart notices.

Part 2

Task Summary (OSINT):

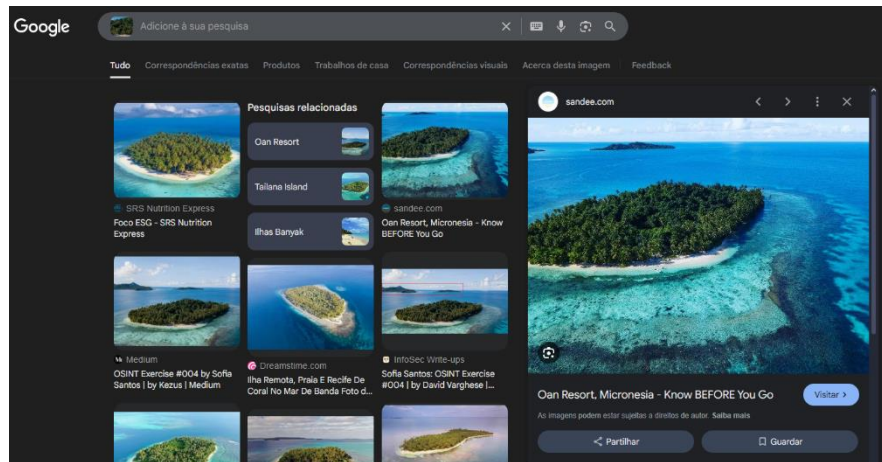
This is a photo of a resort located on an island. To perform this task, it is necessary for the trainees to explain how they arrived at the answers through print screens.

- a) What is the name of the resort?
- b) What are the coordinates of the island?
- c) In which cardinal direction was the camera facing when the photo was taken?

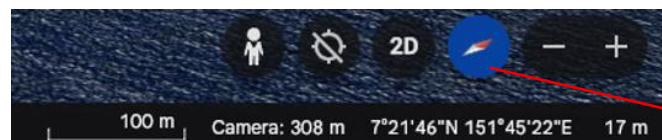
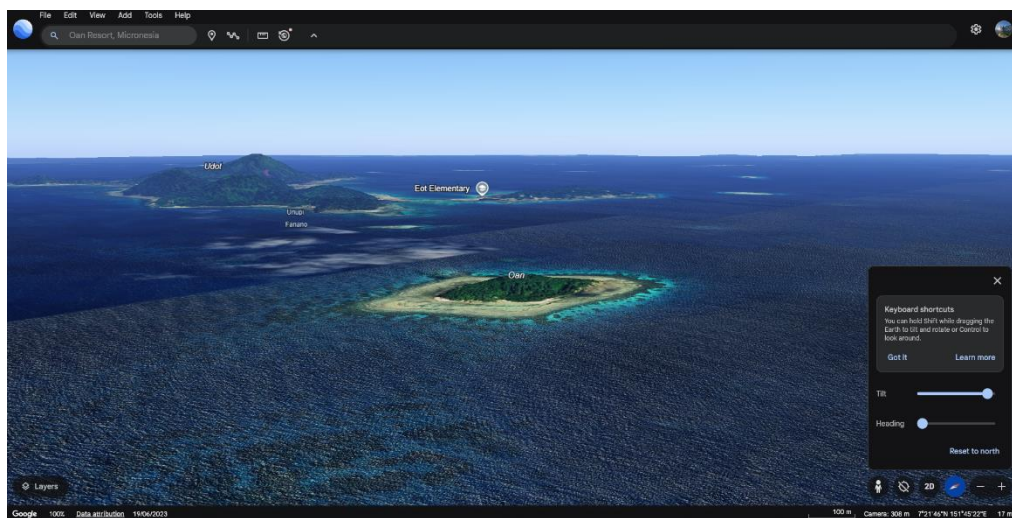


Part 2

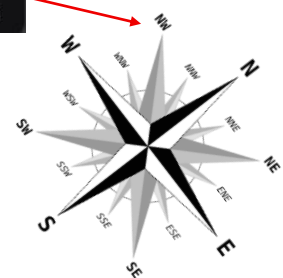
To find out the exact location of the island I simply used reverse image search, eventually finding some results.



With the name of the island (Oan Resort) I can use google earth to see its coordinates and if I experiment with the camera options such as tilt, I get an angle similar to the image provided.



Coordinates: 7°21'46"N 151°45'22"E Cardinal Direction: NW



Part 3

Task Summary (OSINT)

This is a purposeful site for information gathering and hacking.

[Hack This Site](#)

According to what was done in class, it is requested that as much information as possible be collected about the target, which is the site above.

Everything they collect must be subject to proof, it can be through print screens. (IP's, Domains, Emails, Geolocation, etc.)



Hack This Site (TOR .onion HTTPS - HTTP) - IRC - Discord - Forums - Store - URL Shortener - CryptoPaste — Like Us - Follow Us - Fork Us

hackthissite.org

Support HackThisSite
ADVERTISE WITH US

Information wants to be free.

Login (or Register):

Lost your password?
Forgot your username?

Donate

DONATE

basic attention token

HTS costs up to \$300 a month to operate. We need your help!

Challenges

- Basic
- Realistic
- Application
- Programming
- Javascript
- Forensic
- Extended Basic
- Steganography

Training the hacker underground

HackThisSite.org is a free, safe and legal training ground for hackers to test and expand their ethical hacking skills with challenges, CTFs, and more. Active since 2003, we are more than just another hacker wargames site. We are a living, breathing community devoted to learning and sharing ethical hacking knowledge, technical hobbies, programming expertise, with many active projects in development. Join our IRC, Discord, and our forums where users can discuss hacking, network security, and more. Tune in to the hacker underground and get involved with the project!

First timers should read the [HTS Project Guide](#) and [create an account](#) to get started. All users are also required to read and adhere to our [Terms and Conditions](#).

Get involved on our [IRC server](#) (irc.hackthissite.org SSL port 7000), [Discord](#), and our [web forums](#).

LAST CHALLENGE COMPLETED:
» Basic 2 completed 1 minute 51 seconds ago by [\[login to see this feature\]](#)

STAFF BLOGS / SHORT NEWS:

- [blog Status Pages 2.0 - N...](#)
- [news Application 2](#)
- [news Application 7 & ...](#)
- [blog Github for Status Pa...](#)
- [blog Internetswache CTF 20...](#)

LATEST ARTICLES:

- How are websites defaced?
- What is OSINT and what are t...
- How to stay anonymous &a...
- Realistic mission 1 tutorial
- Cryptanalysis

HackThisSite on GitHub:

Repositories	17
Forks	107
Watchers	374
Stargazers	374
Contributors	0
Commits This Week	0

CONTRIBUTE:

- » IRC / Discord (1446 online, 0 voice) - Chat
- » Forums - Discussion
- » GitHub - Open Source

Part 3

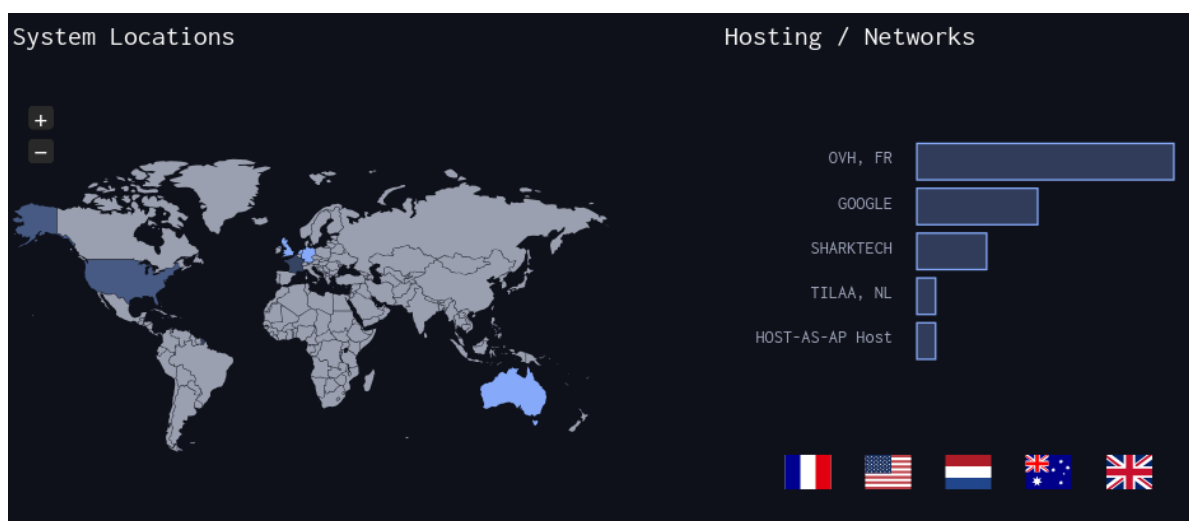
To find the following information I used resources such as: maltego, dns dumpster, netcraft, wappalyzer, cmd (ping, nslookup), page source.

IP: 137.74.187.103

Server IP: 192.168.31.2

```
(joao@kali)-[~]
$ ping www.hackthissite.org
PING www.hackthissite.org (137.74.187.103) 56(84) bytes of data.
```

```
(joao@kali)-[~]
$ nslookup hackthissite.org
Server:      192.168.31.2
Address:     192.168.31.2#53
Non-authoritative answer:
Name:   hackthissite.org
Address: 137.74.187.103
Name:   hackthissite.org
Address: 137.74.187.100
Name:   hackthissite.org
Address: 137.74.187.101
Name:   hackthissite.org
Address: 137.74.187.104
Name:   hackthissite.org
Address: 137.74.187.102
```



Languages / technologies: html, css, rss, xml, javascript, jquery, ssl, php, google webmaster tools, xhtml, open graph, pwa

```
<link href="https://data.htscdn.org/themes/Dark/Dark.css" rel="stylesheet" type="text/css" />
<link href="https://www.hackthissite.org/pages/hts.rss.php" rel="alternate" type="application/rss+xml" title="HTS RSS feed" />
<base href="https://www.hackthissite.org" />
<script type="text/javascript" src="https://data.htscdn.org/js/jquery-1.8.1.min.js"></script>
<script type="text/javascript" src="https://data.htscdn.org/js/notice.min.js"></script>
```

Network

Site	http://www.hackthissite.org	Domain	hackthissite.org
Netblock Owner	OVH Static IP	Nameserver	c.ns.buddyns.com
Hosting company	OVH	Domain registrar	plr.org
Hosting country	 NL	Nameserver organisation	whois.domrobot.com
IPv4 address	137.74.187.101 (VirusTotal)	Organisation	Data Protected, REDACTED, REDACTED, REDACTED, US
IPv4 autonomous systems	AS16276	DNS admin	admin@hackthissite.org
IPv6 address	Not Present	Top Level Domain	Organization entities (.org)
IPv6 autonomous systems	Not Present	DNS Security Extensions	Enabled
Reverse DNS	hackthissite.org		


Wappalyzer

TECHNOLOGIES
MORE INFO
Export

Miscellaneous

-  [Open Graph](#)
-  [PWA](#)
-  [RSS](#)

JavaScript libraries

-  [jQuery](#) 1.8.1

Some Sub-Domains:

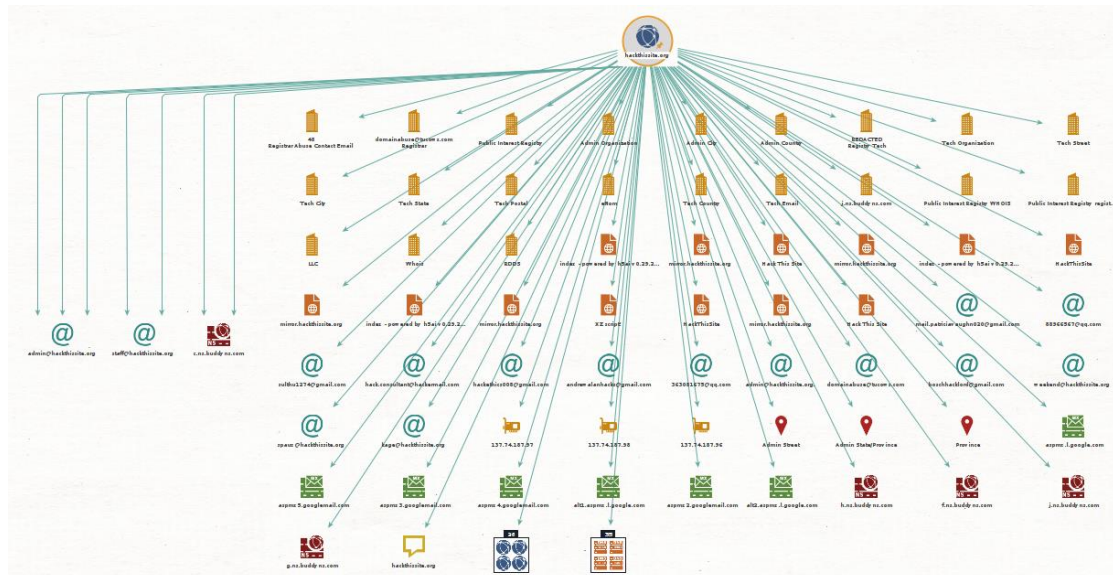
Host	IP	ASN	ASN Name	Open Services (from DB)	RevIP
api.hackthissite.org	137.74.187.116	asn:16276	OWH, FR France		1
vm-005.outbound.firewall.hackthissite.org	137.74.187.154	asn:16276	OWH, FR France		2
vm-050.outbound.firewall.hackthissite.org	137.74.187.155	asn:16276	OWH, FR France		2
vm-099.outbound.firewall.hackthissite.org	137.74.187.159	asn:16276	OWH, FR France		2
vm-150.outbound.firewall.hackthissite.org	137.74.187.158	asn:16276	OWH, FR France		2
vm-200.outbound.firewall.hackthissite.org	137.74.187.157	asn:16276	OWH, FR France		2
git.hackthissite.org	137.74.187.145	asn:16276	OWH, FR France		2
irc.hackthissite.org	137.74.187.150	asn:16276	OWH, FR France	https://www.hackthissite.org/irc/ unknown server HackThisSite IRC	2
irc-hub.hackthissite.org	198.148.81.169	asn:46844	SHARKTECH United States	https://www.irc.hackthissite.org/	1
lille.irc.hackthissite.org	137.74.187.150	asn:16276	OWH, FR France	https://www.irc.hackthissite.org/ unknown server HackThisSite IRC	2
wolf.irc.hackthissite.org	185.24.222.13	asn:196752	TELAA, NL The Netherlands	https://www.irc.hackthissite.org/ nginx/1.18.0 (Ubuntu) 301 Moved Permanently nginx/1.18.0 (Ubuntu) 301 Moved Permanently	2

Discovered through Maltego: ips, emails, different domains, locations, dns names, documents, records

IPS: 137.74.187.103, 137.74.187.97, 137.74.187.98, 137.74.187.96

Emails: admin@hackthissite.org, staff@hackthissite.org, spaux@hackthissite.org, kage@hackthissite.org, _mail.patriciaavaughn020@gmail.com, _88966567@qq.com, weekend@hackthissite.org, _boschhacklord@gmail.com, _domainabuse@ducows.com, 363081675@qq.com, andrewallanhacks@gmail.com, hackethics008@gmail.com, hack.consultant@hackermail.com, sulthu1274@gmail.com

Locations: Netherlands (hosting country), Admin Street, Province, Admin State/Province



Domain (26)		
Entity		
aspmx.googlemail.com	100	
spf.hackmail.org	100	
hackthissite.ca	0	
hackthissite.co	0	
hackthissite.co.uk	0	
hackthissite.com	0	
hackthissite.com.ph	0	
hackthissite.com.ws	0	
hackthissite.edu.ee	0	
hackthissite.edu.ws	0	
hackthissite.gov.mu	0	
hackthissite.gq	0	

DNS ... (35/35)		
Entity		
0-v3stage-cdn.hackthissite.org	100	
1-v3dev-cdn.hackthissite.org	100	
1-v3stage-cdn.hackthissite.org	100	
2-v3stage-cdn.hackthissite.org	100	
adminest.hackthissite.org	100	
forum.hackthissite.org	100	
h5ai.hackthissite.org	100	
hackthissite.org	100	
hp.hackthissite.org	100	
hub.irc.hackthissite.org	100	
irc-services.hackthissite.org	100	
irc-v6.hackthissite.org	100	